

Assignment #2 | Interfaces

PART I: NAME THAT INTERFACE

For this assignment, choose five (5) distinct, real-life examples of the interface types discussed in Chapter 7 (Section 7.2). For each example, answer the following questions:

WIMP (windows, icons, menus, pointer)

Appliance

1) What is the interface, **and** to which interface type does it mostly closely align?

The Honeywell QuietSet 8-Speed Fan is an example of an appliance interface type. An **appliance** interface type is used for a specific task, usually in a short period of time.

2) What is this interface designed to do (ie, what does it help the end user do)?

The Honeywell fan is used to cool down a whole room at 8 cooling and oscillation settings.

3) Which components of WIMP are part of this interface?

The fan features Icons and Menus. The fan's face features 5 icon buttons (power, auto-off timer, cooling setting, oscillation setting, and dimming option). Though it may not be as obvious, there are two main menus. Clicking through the 8 cooling settings by pressing the cooling setting button and clicking through the 4 auto-off timer options by pressing the auto-off timer button.

Touchless

1) What is the interface, **and** to which interface type does it mostly closely align?

The 2010 sports video game Kinect Adventures!, developed by Xbox 360, is an example of the **touchless** interface type.

2) What is this interface designed to do (ie, what does it help the end user do)?

Using gestures and full body motion, players can engage in a variety of minigames. Players' actions are communicated through Kinect's motion camera.

3) Which components of WIMP are part of this interface?

WIMP components include Icon, Menu, and Pointer. Kinect Adventures uses tool icons (ex. settings) and status icons (ex. loading page), a menu option to select the desired activity, and a pointer (which is the user's hand) to select options.

Graphical

1) What is the interface, **and** to which interface type does it mostly closely align?

Pinterest, the social media service that displays visual information as digital pinboards, is an example of a graphical interface type.

2) What is this interface designed to do (ie, what does it help the end user do)?

Pinterest allows users to browse through images based on specific searches (recipes, home decor, style inspiration) or scroll through their customized home pages (algorithms). Users can also save (or “pin”) desired images in specific digital boards (pinboards).

3) Which components of WIMP are part of this interface?

In the web browser version of Pinterest, all WIMP components are used. In the window, users can scroll (touchpad, mouse, or arrow keys), click, or hover. Icons are throughout the interface, by hovering over the icons on the left sidebar, the application will display an icon description (ex. Create, Messages, Updates, Settings & Support). Menus are also throughout the interface. The search bar displays a “Recent searches” menu, at the top of the home page there’s a scrollable menu with the titles of the user’s pinboards, and the navigation icon on the left sidebar displays common categories (ex. Beauty, Design, Food & Drinks). Unlike the mobile version of Pinterest, the web browser has a pointer to easily navigate through the application.

Wearable

1) What is the interface, **and** to which interface type does it mostly closely align?

The clothing brand Stone Island has created a collection of thermosensitive outerwear. This heat sensitive clothing best aligns with the **wearable** interface type. Wearable is a broad category that describes devices that are worn on the body.

2) What is this interface designed to do (ie, what does it help the end user do)?

Stone Island’s outerwear can communicate to the wearer (user) through temperature-induced changes. The outerwear is sensitive to cold and hot temperatures and the wearer’s behavior.

3) Which components of WIMP are part of this interface?

Not applicable.

Smart

1) What is the interface, **and** to which interface type does it mostly closely align?

The Amazon Echo Pop Smart Speaker with Alexa is an example of a **smart** interface type.

2) What is this interface designed to do (ie, what does it help the end user do)?

The Amazon speaker allows users to enjoy music, audiobooks, and podcasts via Bluetooth throughout the home, utilize voice commands with the virtual assistant Alexa, control other

compatible smart home devices with built-in motion or indoor temperature sensors, and convert the home into a musical concert or home theater using compatible Echo devices in different rooms or the Fire TV.

3) Which components of WIMP are part of this interface?

The Amazon speaker has one WIMP component, Icon. On the body of the speaker, there are four primary icon buttons: action, volume up, volume down, and mic off. The action button includes saying the “wake up” word, silencing alarms, putting the device into setup mode, and more.

PART II: WHAT’S IN AN INTERFACE?

Think about what a "good" and "bad" interface looks like. What is an example of a "good" interface? What makes it good? Now think about a "bad" interface. What makes it bad? In your rationale for both examples, be sure to discuss the usability of relevant parts of the interface (eg, WIMP), what features of the interface make it a "good" or "bad", and how the interface impacts the user experience. You can use any of the examples that you identified above or find different examples.

Good Interface (Pinterest): My favorite social media application is Pinterest, and for good reason! Though Pinterest is primarily for saving and sharing images, there are many features that are customizable, giving the user control.

- Minimalistic/slick icon design allows for images to speak for themselves (not distracting)
- Readable font choice
- Multiple ways to perform the same action (ex. Using the arrow keys or touchpad to scroll)
- Very little color makes it readable for visually-impaired (black text, white background, red Pinterest logo)
- Conceptual modeling + logos (magnifying glass for search or house for home page)
- Hovering pointer over images affords more actions (saving image to specific board, visiting the link, or uploading image to another social media platform)
- Though the target audience are 18-34 year old women, people of older ages can use the application. Pinterest is very intuitive.

Bad Interface (Shein Website Home Page): As much as I understand the convenience of online shopping, I rarely shop because of the endless options! Even with filters, categories, and low prices, there are many overwhelming aspects that prevent me from purchasing.

- Cluttered home page
- Extensive drop-down category menu
- Side bar with discount code
- Pop-up ads!

- Overuse of icons under each clothing item (marketed discount percentage, price, add-to-cart button, clock (for limited time only price?))
- Search bar has 10 Recently Searched items and 10 Search Discovery items (too much)
- Endless scrolling (able to view various items just on the home page (jewelry, clothing, hair accessories))
- Complex navigation - There are multiple ways to accomplish the same task (are there *too* many ways?)

PART III: PRACTICING GESTALT

Look at each of the stimuli **(a) through (e)** in the below image. Using the Gestalt Laws of perceptual organization, describe, in your own words, what "patterns" or organizing principles are displayed.

A - Proximity

This stimulus displays three groupings of circles, one grouping has three shaded circles in a linear formation, another grouping has three shaded circles in a triangular formation, and the last grouping has two shaded circles in a linear formation. Visually, the separation between the groupings and closeness of the circles within each grouping signal a connection. The separation also breaks the content into “digestible chunks.”

B - Common Region

This stimulus displays three small, medium, and large shaded triangles with three small, medium, and large shaded circles. The six shapes are confined in a specific boundary, with associated groupings based on size. For example, the small triangle and small circle are close together, as well as the medium triangle and medium circle, and large triangle and medium circle.

C - Closure

This stimulus displays a circular shape that isn't completely connected. Closure is the best pattern because the user naturally completes the partially connected circle from familiar patterns.

D - Continuity

This stimulus displays 9 small shaded circles in a diagonal formation and 5 large shaded circles in a curved cross-section across the small circles. The users' eyes naturally follow the circles' paths and the users' mind associates the circles' arrangements as continuous lines and curves.

E - Symmetry & Order

This stimulus displays five different shapes, each shape divided in half (either vertically or horizontally). Each shape has an equal distance from the following shape, creating a balance between the assortment of shapes. Also, placing a divided rhombus in the center avoids “visual heaviness” on either side.